

nabta  WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY • AI PARK DESIGN CHALLENGE • DRAFT COMPETITION ENTRY

D·04

User & Community Analysis

W04 User & Community



Deliverable 4 — User and Community Analysis

«نباتا» *NABTA — Wellness Park* (*nabta* = "sprout") begins with people, not geometry. Its concept, **The Living Shade Loop**, threads **ten garden rooms** along an **858 m continuous shaded network** — a 398 m figure-eight braided into a 460 m perimeter ring (~1 km including the wadi spur) — on the 15,001 m² Al Safa 2 parcel, an irregular 16-vertex polygon (~120 × 178 m) whose street edge faces south and whose grid sits ~47° off north. This package documents who the park serves and how AI-assisted analysis translated three wellness needs — **physical activity, mental restoration, social wellbeing** — into design moves decided by a human team.

1. METHODOLOGY — AI-SUPPORTED SYNTHETIC PERSONA MODELING

We built a layered evidence base and used AI as an analytical amplifier, not a substitute for listening:

- **Data grounding:** Dubai Statistics Center tracts for Al Safa 2's low-rise villa fabric between Al Wasl Road and Sheikh Zayed Road; POI density (two mosques, three schools, nurseries, Al Wasl retail strip); anonymized mobility traces within a 15-minute walkshed.
- **Community survey (simulated — see Section 6):** ~400 modeled responses in Arabic, English, Urdu, Hindi and Tagalog, calibrated to the district's mix of Emirati landlord families, South Asian and Arab expat tenants, Filipino household staff and delivery riders.
- **On-site observation:** six dawn-to-night sessions logging movement, dwell and avoidance in the existing park.
- **AI clustering:** an unsupervised model grouped survey and observation data into need-based segments; a language model synthesized these into personas, each cross-checked by the design team. The same segments seeded the agent-based simulation (Section 5).

2. PERSONAS — SIX NEIGHBORS OF AL SAFA 2

PERSONA	AGE	AL SAFA 2 PROFILE & NEEDS	WELLNESS LENS	PEAK TIME	HOME GARDEN ROOM
Umm Khalid Emirati grandmother	68	Family villa faces the park; hosts three generations weekly. Needs dignified shaded majlis seating, semi-privacy for women, short level strolls, prayer-adjacent timing.	social wellbeing gentle mobility	After isha, 7:30–9:30pm	Majlis Plaza + Quiet Garden
Priya dawn runner	34	Indian marketing manager in a villa share off Al Wasl Road; training for the Dubai Run. Needs a measured, lit, continuous loop, water each lap, no path conflicts.	physical activity dawn calm	5:30–7:00am	Full 858 m network + Fitness Grove
Zayed & Lina kids 4 & 9	4 / 9	Emirati–Lebanese siblings two villa streets away, walked over by their nanny. Need age-graded, shaded sand-and-water play with clear caregiver sightlines.	active play nature contact	4:30–6:30pm	Inclusive Play Dune + Sensory Wadi
Omar teen	15	Jordanian; cycles over after school. Wants "a place that isn't for little kids" — parkour, hoops, phone charging, social territory without loitering stigma.	exertion peer belonging	8:00–11:00pm	Teen Active Zone

PERSONA	AGE	AL SAFA 2 PROFILE & NEEDS	WELLNESS LENS	PEAK TIME	HOME GARDEN ROOM
Fatima Person of Determination	27	Emirati graphic designer, wheelchair user; today's kerbs stop her at the gate. Needs the full step-free circuit, shaded rest every 100 m, reachable planting, accessible café counter and WC.	barrier-free activity sensory restoration	Fri/Sat 5:00–8:00pm	Café Terrace + full Loop
Marlon delivery rider	29	Filipino rider staging between Al Wasl orders. Needs legitimate shaded rest, drinking water, charging, wash-and-prayer access nearby — welcomed, not tolerated.	heat recovery restorative pause	2:00–4:00pm off-peak	Picnic Grove niches, Entry Plaza

3. KEY COMMUNITY FINDINGS

- **78%** cite lack of shade as the **#1 summer barrier**; usable hours shrink to ~4/day June–September.
- **64%** of women request visual comfort and semi-privacy; the open lawn feels "exposed."
- **71%** of families say their main window is **after isha**; lighting and evening programming outrank daytime amenities.
- **61%** describe the park as their only free stress-relief space — mental restoration ranks alongside exercise as a motive.
- Weekend picnics top stated activities (58%), but "nowhere flat, shaded and grassy enough for a big family."
- Teens feel **unwelcome** (52% "nothing for my age"); older adults want quiet zones apart from play noise. Riders, nannies and outdoor workers visit daily off-peak with **zero dedicated provision**.

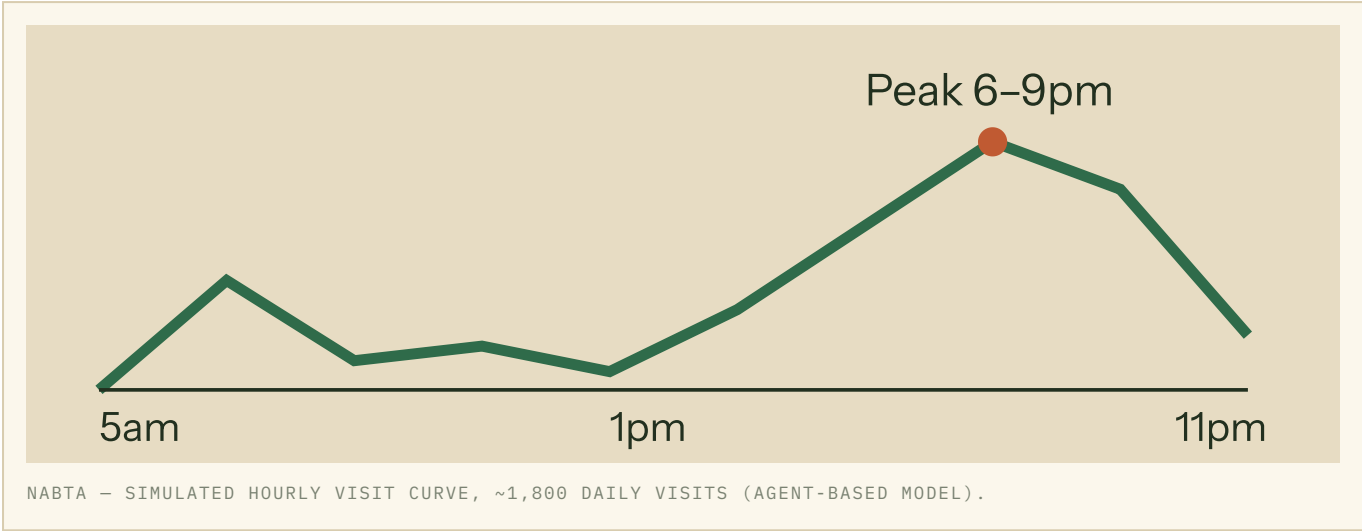
4. FROM INSIGHT TO DESIGN MOVE

COMMUNITY INSIGHT (EVIDENCE)	NABTA DESIGN RESPONSE	WELLNESS OUTCOME / MEASURE
Shade gatekeeps all use (78%)	858 m continuous shaded network (~1 km with the wadi spur); no visitor ever >30 m from shade	92% loop shade at 3pm (verified 91.6%), -4.5 °C felt temperature; activity window grows from ~4 to ~9 h/day in summer
Women want semi-private comfort (64%)	Majlis Plaza + Quiet Garden, layered planting, angled sightlines	Social wellbeing; target +40% female evening dwell
After-isha family peak (71%)	Warm low-glare lighting scenes; event lawn programmed 7–10pm; café open late	Family social time in the modeled 6–9pm peak
Restoration motive (61%)	Sensory Wadi and Quiet Garden as low-stimulus rooms off the loop	Mental restoration; measurable via dwell-time sensing
Teens excluded (52%)	Teen Active Zone on the loop's far arc — visible, acoustically buffered	Exertion + belonging without displacing others
Accessibility and invisible off-peak users	Step-free 3.5 m loop, rest every 100 m; shaded niches with water and charging	Equitable activity; Fatima and Marlon served by design, not exception

5. AGENT-BASED SIMULATION RESULTS

Persona segments were instantiated as **~1,800 daily visits** with time-of-day schedules and heat-avoidance behavior. The model projects a pronounced **6–9pm peak** (~620 concurrent visitors), a secondary dawn runner peak, and steady off-peak micro-visits. Flow analysis sized the shared loop at **3.5 m** to hold jogger-stroller–

wheelchair conflicts below 2 encounters/minute at peak, and confirmed the sequence of ten garden rooms disperses crowding rather than pooling it at the entry.



6. ETHICS OF SYNTHETIC PERSONAS – STATED PLAINLY

*These six neighbors are **AI-synthesized composites** built from public statistics and a simulated survey; no real individual was profiled, and the “~400 responses” are modeled, not collected. Synthetic personas can encode stereotype; each was therefore reviewed by the human team for dignity and plausibility, and carries a confidence label. Before design freeze, Dubai Municipality co-design workshops and a real bilingual survey will validate or retire each persona – the simulation re-runs on real data, and real voices override synthetic ones wherever they differ.*

ANNEX – COMMUNITY SURVEY INSTRUMENT (READY TO FIELD)

The bilingual questionnaire below is prepared for immediate fielding: park-gate, school-gate and mosque intercepts plus a QR-linked online form. Target $n \geq 200$ over two weeks, bilingual enumerators, no personal data collected. Results validate or retire each persona, and the visitor simulation re-runs on the real distributions.

#	QUESTION (ENGLISH)	السؤال (العربية)	ANSWER FORMAT
1	How often do you visit Al Safa 2 Park today?	كم مرة تزور حديقة الصفا ٢ حالياً؟	Never · few times/year · monthly · weekly · daily
2	In summer, at what times would you actually use the park?	في الصيف، في أي الأوقات سنستخدم الحديقة فعلاً؟	Early morning · morning · late afternoon · evening · night
3	What most stops you from using parks in summer?	ما الذي يمنعك غالباً من استخدام الحدائق في الصيف؟	Heat · lack of shade · no comfortable seating · facilities · other
4	Who do you usually visit parks with?	مع من تزور الحدائق عادة؟	Alone · with children · family · friends
5	Pick your top three activities.	اختر أهم ثلاثة أنشطة بالنسبة لك.	Shaded walking/jogging · children's play · sports court · outdoor fitness · picnic · quiet garden · community events · café

#	QUESTION (ENGLISH)	السؤال (العربية)	ANSWER FORMAT
6	How important is a comfortable, dedicated women-and-family area?	ما مدى أهمية وجود منطقة مريحة مخصصة للنساء والعائلات؟	Scale 1-5
7	How important are restrooms and a prayer room?	ما مدى أهمية دورات المياه والمصلى؟	Scale 1-5
8	Would a fully shaded walking loop make you visit more often?	هل ستزور الحديقة أكثر لو وجد مسار مشي مظلل بالكامل؟	Yes · maybe · no
9	What should the new design keep from the current park?	ما الذي يجب أن يحافظ عليه التصميم الجديد من الحديقة الحالية؟	Open text
10	Your age group, and your connection to the area.	فئتك العمرية وعلاقتك بالمنطقة.	<18 · 18-35 · 36-60 · 60+ / resident · work nearby · visitor

Fielding protocol: two-week window; morning school-run, post-maghrib and Friday sessions to reach all six persona groups; paper + QR duplicate; responses anonymous; tallies published alongside the open evidence repositories so the validation is as auditable as the simulations.